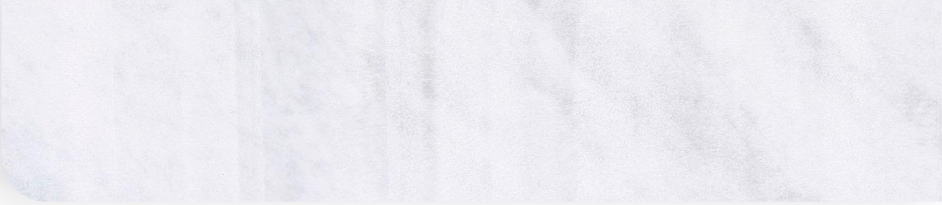


Module 4

Morals, values and Ethics - Integrity- Academic integrity - Work Ethics
- Service Learning - Civic Virtue - Respect for others - Living
peacefully - Caring and Sharing - Honesty - Courage - Cooperation
commitment - Empathy - Self Confidence - Social Expectations
Case study on Engineering disasters (include study of ethical issues in a
recent disaster)
Code of Conduct in Engineering profession

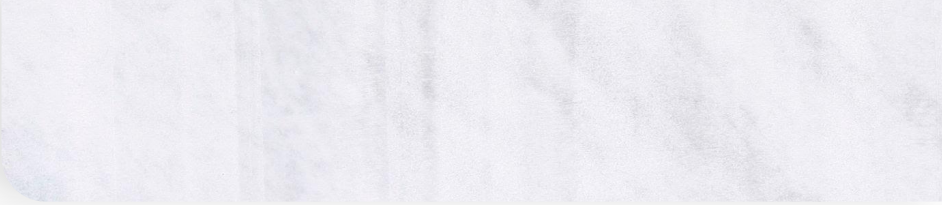




Morality

- Refers to the set of **principles, norms, and standards of right and wrong behavior** accepted by society or an individual.
- It is a **practical application** of morals in daily life.
- Morality helps us decide “*how we should act*” in a given situation.
- In professional ethics, morality means making **decisions that align with society’s welfare**.
- Example:
 - An engineer refusing to approve a faulty design → moral responsibility.
 - A doctor treating a poor patient without fee in emergency → moral action.

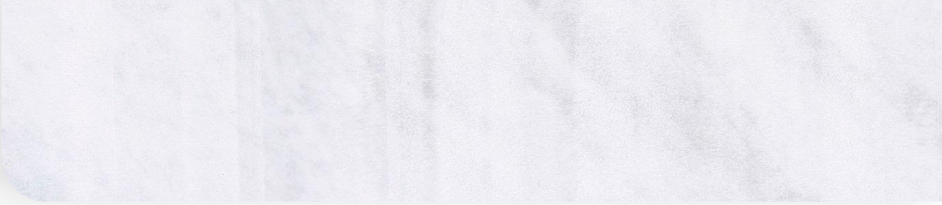




Importance of Morality

- Builds **trust and respect** in society.
- Promotes **peaceful coexistence**.
- Prevents **corruption, exploitation, and injustice**.
- Helps professionals take **ethical decisions** when rules are unclear.





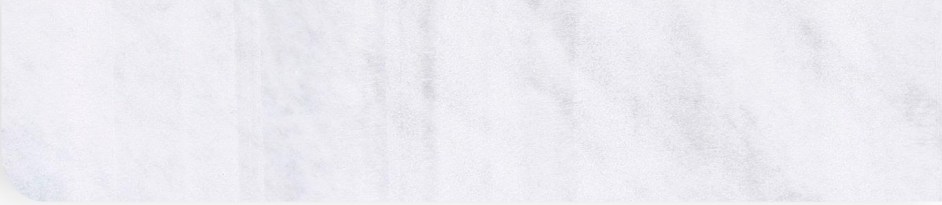
Moral Dilemma

- A **Moral Dilemma** is a situation where a person faces a **conflict between two or more moral principles or values**, and **choosing one option leads to the violation of another**.
- It is often described as a “*no-win situation*” where **every choice has both ethical and unethical consequences**.

Characteristics of a Moral Dilemma

1. **Conflict of Values** → Two or more moral principles come into conflict.
2. **No Clear Solution** → Whichever choice is made, some moral rule is compromised.

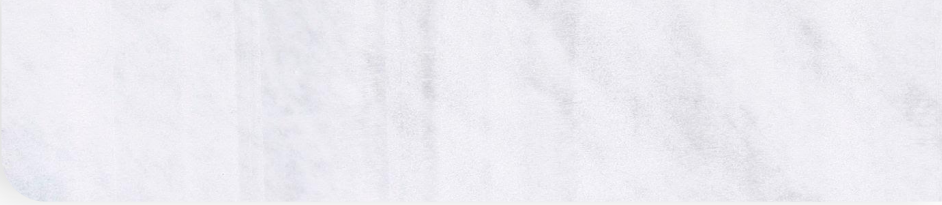




3. **Emotional and Ethical Pressure** → The decision often involves guilt, responsibility, or social consequences.

4. **Real-Life Relevance** → Common in professions like engineering, medicine, law, and business.





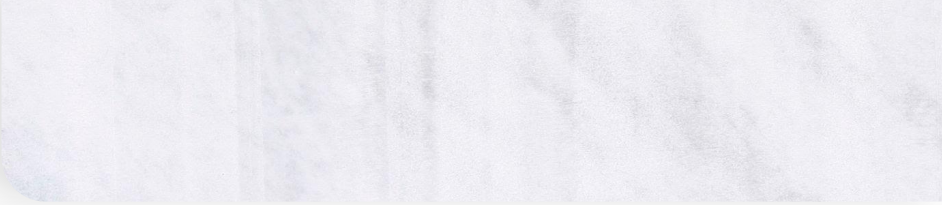
Values

- Values are the **deeply held beliefs and ideals** that guide an individual's behavior and decision-making.
- They reflect “*what is important in life*”.
- Values are more **universal** than morals but vary from person to person.

Examples of Values:

- Honesty → Being truthful in all actions.
 - Respect → Treating others with dignity.
 - Responsibility → Owning one's duties.
 - Hard Work → Believing in effort and perseverance.
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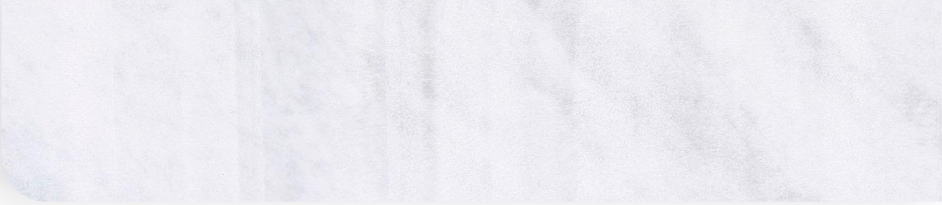
Ethics

- Ethics are the **systematic rules and codes of conduct** that govern **professional and social behavior**.
- Ethics provide a **framework** for resolving dilemmas where morals and values may conflict.
- They are often **codified** in professional codes (like medical ethics, engineering ethics).

Examples of Ethics:

- An engineer must ensure public safety before profit.
 - A doctor must maintain patient confidentiality.
 - A teacher must avoid favoritism and treat all students fairly.
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INTEGRITY

- Defined as the unity of thought, word and deed (honesty) and open mindedness.
- It is used to describe a person's level of honesty, commitment or willingness to do what is right
- Honesty and integrity are close to each other
- Integrity leads to consistency of character, decision and action and helps to gain confidence.
- It enthuse people not only to execute a job well but to achieve excellence in performance.
- It helps them to own the responsibility and earn self-respect and recognition by doing the job



Academic Integrity

- Academic integrity is the commitment to and demonstration of honest and moral behaviour in an academic setting.
- This is most relevant at the university level as it relates to providing credit to other people when using their ideas.

International Centre for Academic Integrity(ICAI) stated 5 pillars of academic integrity:

- 1. Honesty*
- 2. Trust*
- 3. Fairness*



4. Respect/recognition

5. Responsibility

WORK ETHICS

It is defined as a set of attitudes concerned with the value of work, which forms the motivational orientation.

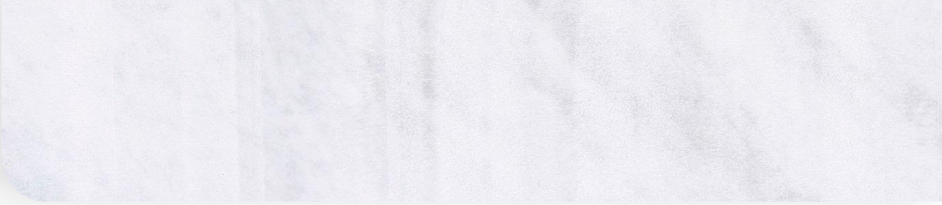
The 'work ethics' is aimed at ensuring:

- the economy (get job, create wealth, earn salary),
- productivity (wealth, profit),
- safety (in workplace),

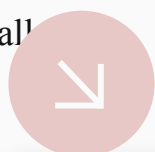


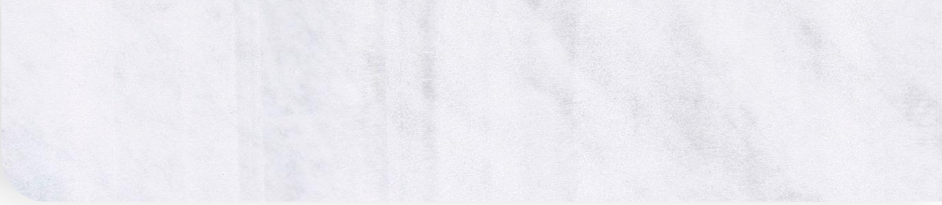
- health and hygiene (working conditions),
- privacy (raise family),
- welfare (social work),
- security (permanence against contractual, pension, and retirement benefits)
- cultural and social development (leisure, hobby, and happiness),
- environment (anti-pollution activities), and offer opportunities for all, according to their abilities, but without discrimination.





SERVICE LEARNING

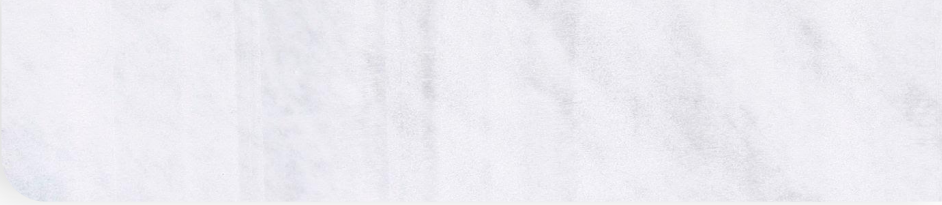
- Service learning refers to learning the service policies, procedures, norms, and conditions, other than ‘the technical trade practices’.
 - It includes the characteristics of the work, basic requirements, security of the job, and awareness of the procedures, while taking decisions and actions.
 - It helps the individuals to interact ethically with colleagues, to effectively coordinate with other departments, to interact cordially with suppliers as well as the customers, and to maintain all friendly interactions.
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VIRTUES

- Virtues are positive and preferred values.
 - Virtues are desirable attitudes or character traits, motives and emotions that enable us to be successful and to act in ways that develop our highest potential.
 - Honesty, courage, compassion, generosity, fidelity, integrity, fairness, transparency, self-control, and prudence are all examples of virtues.
 - They are like habits, once acquired, they become characteristics of a person.
 - The virtuous person is the ethical person.
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Civic Virtues

- Civic virtues are the moral duties and rights, as a citizen of the village or the country or an integral part of the society and environment.
- An individual may exhibit civic virtues by voting, volunteering, and organizing welfare groups and meetings.

The duties are:

1. To pay taxes to the local government and state, in time.
2. To keep the surroundings clean and green.





3. Not to pollute the water, land, and air by following hygiene and proper garbage disposal.

Ex: not to burn wood, tyres, plastic materials, spit in the open, even not to smoke in the open, and not to cause nuisance to the public, are some of the civic (duties) virtues.

4. To follow the road safety rules.

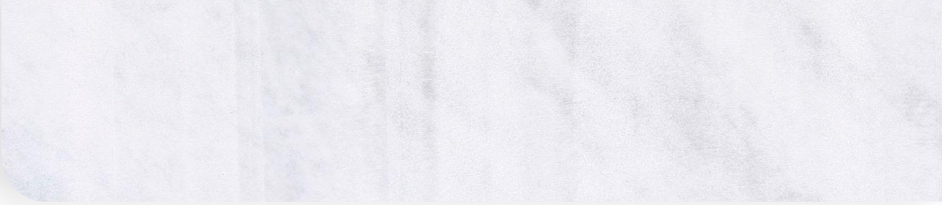
The rights are:

1. To vote the local or state government.

2. To contest in the elections to the local or state government.

3. To seek a public welfare facility such as a school, hospital or a community hall or transport or communication facility, for the residents.



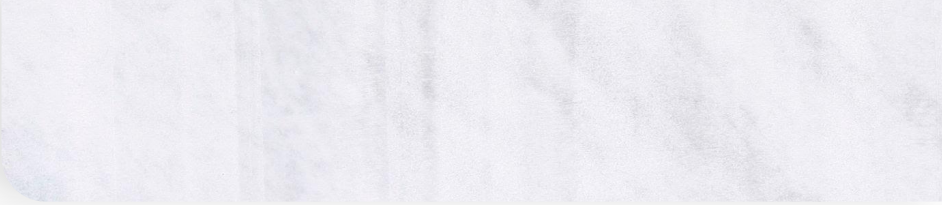


4. To establish a green and safe environment, pollution free, corruption free, and to follow ethical principles. People are said to have the right to breathe in fresh air, by not allowing smoking in public.

5. People have inalienable right to accept or reject a project in their area. One has the right to seek legal remedy, in this respect, through public interest petition.

George Washington 4 embodied the civic virtues as indispensable for a self-governing administration. These virtues are divided into four categories:

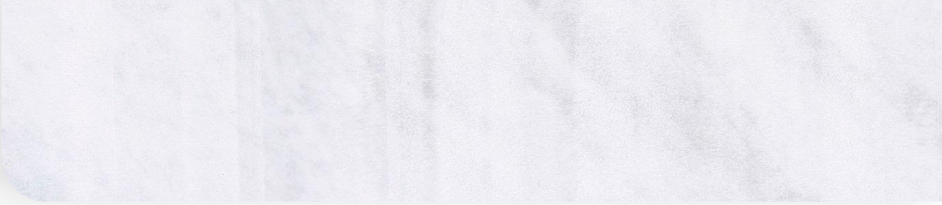




1. Civic Knowledge:

- Citizens must understand what the Constitution says about how the government is working, and what the government is supposed to do and what not to do.
- We must understand the basis of our responsibilities as citizens, besides duties and rights.
- We must be able to recognize when the government or another citizen infringes upon our rights.
- It implies that the government requires the participation of the enlightened citizens, to serve and survive.





2. Self-Restraint:

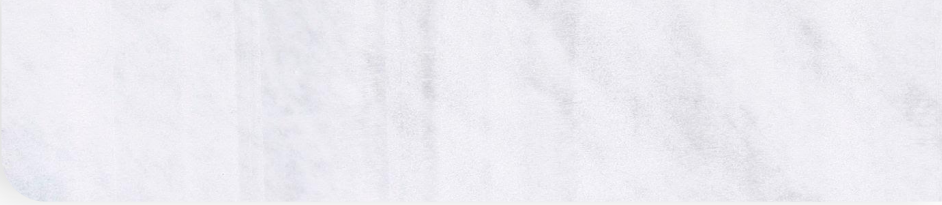
- For citizens to live in a free society with limited government each citizen must be able to control or restrain himself; otherwise, we would need a police state—that is, a dictatorial government to maintain safety and order.
- He advocated and demonstrated self-restraint several times in his private and public life, and naturally he was a great leader.



3. Self-assertion:

- Means that citizens must be proud of their rights, and have the courage to stand up in public and defend their rights.
- Sometimes, a government may usurp the very rights that it was created to protect.
- In such cases, it is the right of the people to alter or abolish that government (e.g., voting rights, rights call back).

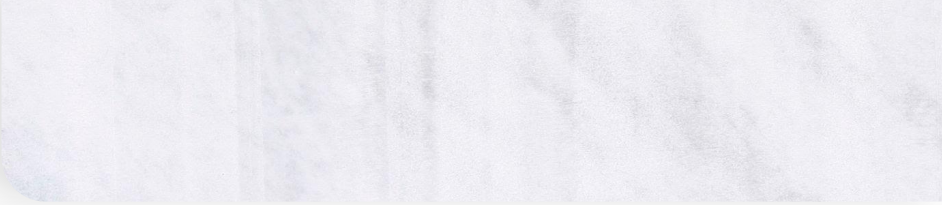




4. Self-Reliance:

- Self-reliant citizens are free citizens in the sense that they are not dependent on others for their basic needs.
- They do not need a large provider-government, which has the potential to become an oppressive government, to meet those needs.
- Only a strong self-reliant citizenry will be able to enjoy fully the blessings of liberty.





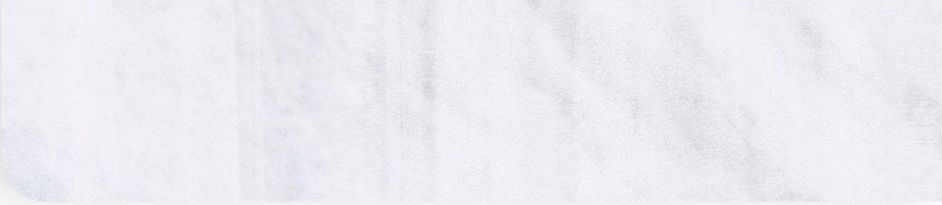
RESPECT FOR OTHERS

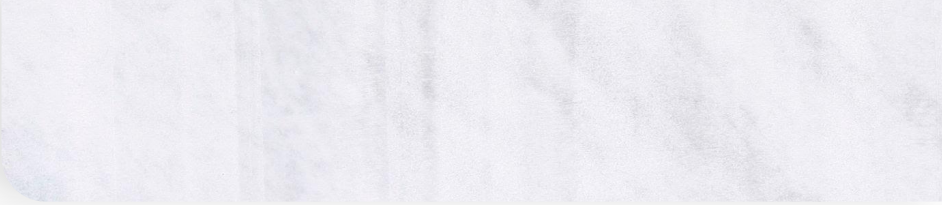
The basic requirement for nurturing friendship, team work, and for the synergy it promotes and sustains.

The principles are:

1. Recognize and accept the existence of other persons as human beings, because they have a right to live, just as you have.
2. Respect others' ideas (decisions), words, and labor (actions).
 - One need not accept or approve or award them, but shall listen to them first.
 - One can correct or warn, if they commit mistakes.



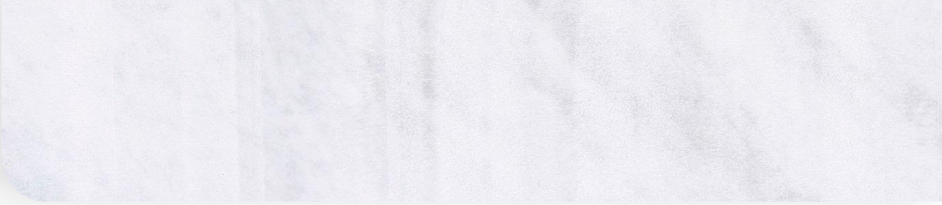
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- Appreciate colleagues and subordinates on their positive actions.
 - Criticize constructively and encourage them. They are bound to improve their performance, by learning properly and by putting more efforts.
3. Show ‘goodwill’ on others. Love others. Allow others to grow.. This will facilitate focus, coherence, and strength to achieve the goals
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LIVING PEACEFULLY

- To live peacefully, one should start install peace within (self).
- Charity begins at home.
- Then one can spread peace to family, organisation where one works, and then to the world, including the environment.
- Only who are at peace can spread peace.
- One should adopt the following means to live peacefully, in the world:





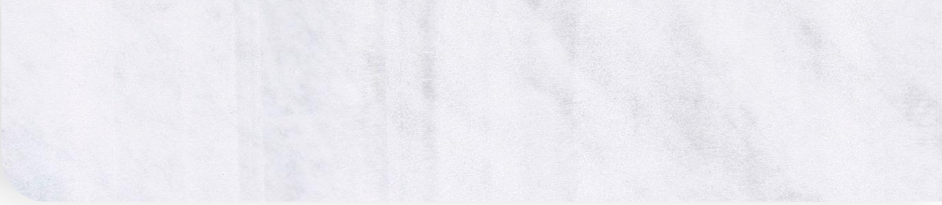
Nurture

1. Order in one's life (self-regulation, discipline, and duty).
2. Pure thoughts in one's soul (loving others, blessing others, friendly, and not criticizing or hurting others by thought, word or deed).
3. Creativity in one's head (useful and constructive).
4. Beauty in one's heart (love, service, happiness, and peace).

Get

5. Good health/body (physical strength for service).





Act

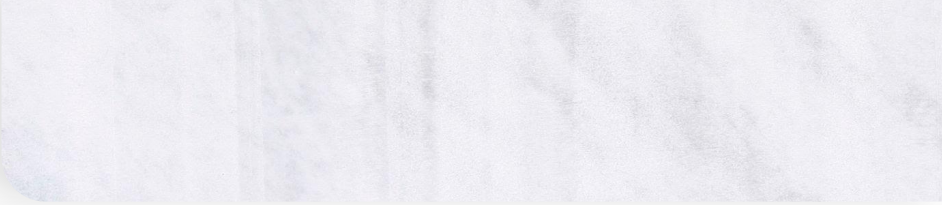
6. Help the needy with head, heart, and hands (charity). Service to the poor is considered holier than the service to God.

7. Not hurting and torturing others either physically, verbally, or mentally.

The following are the factors that promote living, with internal and external peace:

1. Conducive environment (safe, ventilated, illuminated and comfortable).
 2. Secured job and motivated with 'recognition and reward'.
 3. Absence of threat or tension by pressure due to limitations of money or time.
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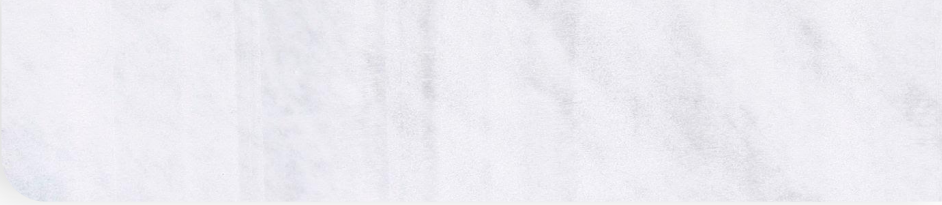


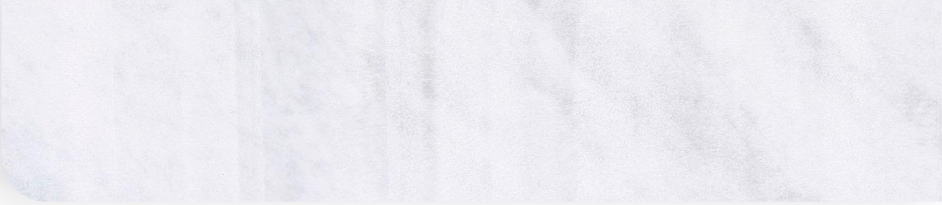
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4. Absence of unnecessary interference or disturbance, except as guidelines.
 5. Healthy labour relations and family situations.
 6. Service to the needy (physically and mentally-challenged) with love and sympathy.

CARING

- Caring is feeling for others.
- In the context of professional ethics caring is a process which exhibits the interest in, and support for, the welfare of others with fairness, impartiality and justice in all activities, among the employees



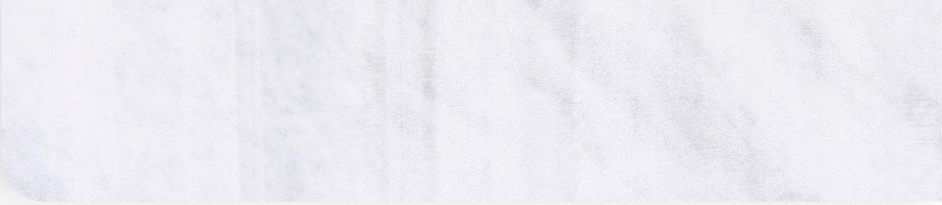
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- It includes showing respect to the feelings of others, and also respecting and preserving the interests of all others concerned.
 - Caring is reflected in activities such as friendship, membership in social clubs and professional societies, and through various transactions in the family, fraternity, community, country and in international councils.
 - In the present day context, caring for the environment (including the fauna and flora) has become a necessity for our very survival.
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SHARING

- Primarily, caring influences ‘sharing’.
- Sharing is a process that describes the transfer of knowledge (teaching, learning, and information), experience (training), commodities (material possession) and facilities with others.
- The transfer should be genuine, legal, positive, voluntary, and without any expectation in return.
- However, the proprietary information it should not be shared with outsiders.
- Through this process of sharing, experience, expertise, wisdom and other benefits reach more people faster.

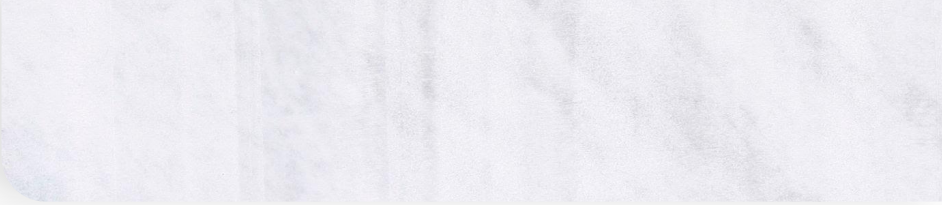


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- Sharing is voluntary and it can not be driven by force, but motivated successfully through ethical principles.
 - In short, sharing is ‘charity’.
 - For the humanity, ‘sharing’ is a culture.
 - The ‘happiness and wealth’ are multiplied and the ‘crimes and sufferings’ are reduced, by sharing.
 - Philosophically, the sharing maximizes the happiness for all the human beings.
 - In terms of psychology, the fear, divide and distrust between the ‘haves’ and ‘have-nots’ disappear.
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HONESTY

- Honesty is a virtue, and it is exhibited in two aspects namely,
 - (a) Truthfulness and
 - (b) Trustworthiness.
- Truthfulness is to face the responsibilities upon telling truth.
- One should keep one's word or promise.
- By admitting one's mistake committed, it is easy to fix them.



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- Reliable engineering judgment, maintenance of truth, defending the truth, and communicating the truth, only when it does ‘good’ to others, are some of the reflections of truthfulness.
 - Trustworthiness is maintaining integrity and taking responsibility for personal performance.
 - Honesty is mirrored in many ways.

The common reflections are:

- (a) Beliefs (intellectual honesty).
- (b) Communication (writing and speech).
- (c) Decisions (ideas, discretion).



(d) Actions (means, timing, place, and the goals). and

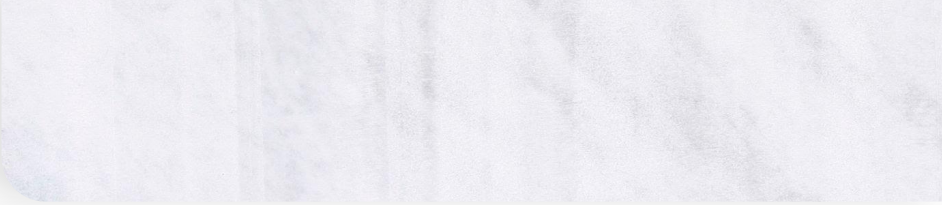
(e) Intended and unintended results achieved.

Some of the actions of an engineer that leads to dishonesty are:

1. Lying:

- Honesty implies avoidance of lying.
- An engineer may communicate wrong or distorted test results intentionally or otherwise.
- It is giving wrong information to the right people





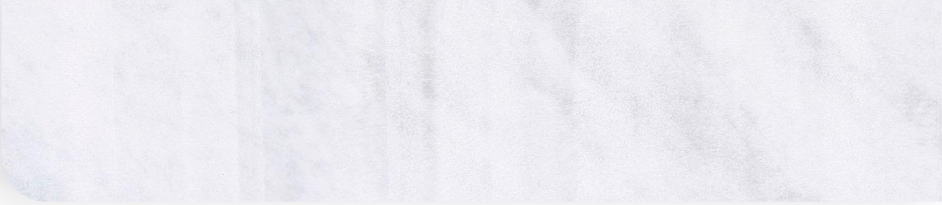
2. Deliberate deception:

- An engineer may judge or decide on matters one is not familiar or with insufficient data or proof, to impress upon the customers or employers.
- This is a self- deceit.

3. Withholding the information:

- It means hiding the facts during communication to one's superior or subordinate, intentionally or otherwise.





4. Not seeking the truth:

Some engineers accept the information or data, without applying their mind and seeking the truth.

5. Not maintaining confidentiality:

- It is giving right information to wrong people.
- The engineers should keep information of their customers/clients or of their employers confidential and should not discuss them with others.

6. Giving professional judgment under the influence of extraneous factors such as personal benefits and prejudice.

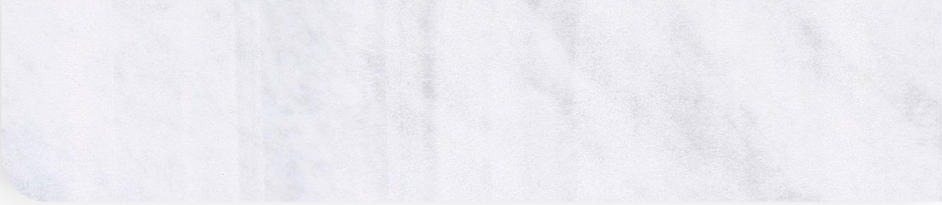


The laws, experience, social welfare, and even conscience are given a go-bye by such actions. Certainly this is a higher-order crime

COURAGE

- Courage is the tendency to accept and face risks and difficult tasks in rational ways.
- Self-confidence is the basic requirement to nurture courage.
- Courage is classified into three types, based on the types of risks, namely
 - (a) Physical courage,
 - (b) Social courage, and
 - (c) Intellectual courage





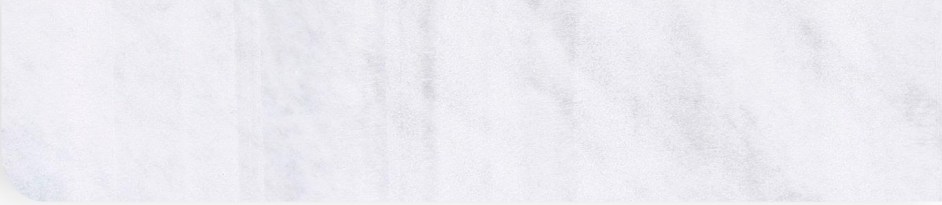
In ***physical courage***, the thrust is on the adequacy of the physical strength, including the muscle power and armaments.

The ***social courage*** involves the decisions and actions to change the order, based on the conviction for or against certain social behaviors.

This requires leadership abilities, including empathy and sacrifice, to mobilize and motivate the followers, for the social cause

The ***intellectual courage*** is inculcated in people through acquired knowledge, experience, games, tactics, education, and training.

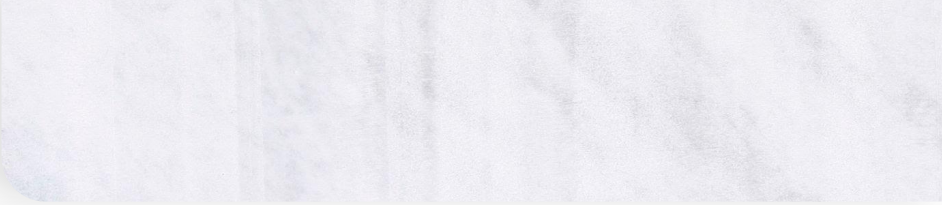




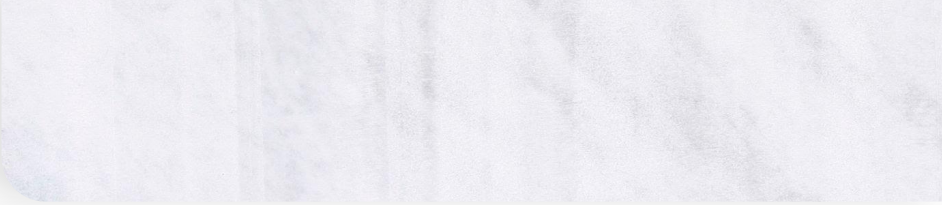
In professional ethics, courage is applicable to the employers, employees, public, and the press. Look before you leap. One should perform Strengths, Weakness, Opportunities, and Threat (SWOT) analysis.

- Calculate (estimate) the risks, compare with one's strengths, and anticipate the end results, while taking decisions and before getting into action.
- Learning from the past helps.
- Past experience and wisdom gained from self-study or others will prepare one to plan and act with self-confidence, succeed in achieving the desired ethical goals through ethical means.



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- Opportunities and threat existing and likely to exist in future are also to be studied and measures to be planned.
 - This will help anyone to face the future with courage.
 - The courageous people own and have shown the following characteristics, in their professions:
 - (a) Perseverance (sustained hard work),
 - (b) Experimentation (preparedness to face challenges, that is, unexpected or unintended results),
 - (c) Involvement (attitude, clear and firm resolve to act), and
 - (d) Commitment (willing to get into action and to reach the desired goals by any alternative but ethical means).
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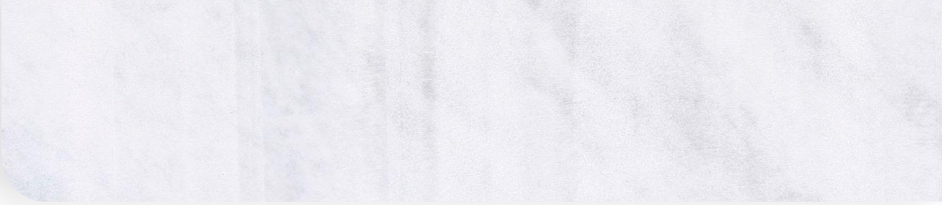




COOPERATION

- It is a team-spirit present with every individual engaged in engineering.
- Co-operation is activity between two persons or sectors that aims at integration of operations (synergy), while not sacrificing the autonomy of either party.
- Further, working together ensures, coherence, i.e., blending of different skills required, towards common goals
- Willingness to understand others, think and act together and putting this into practice, is cooperation





According to professional ethics, cooperation should exist or be developed, and maintained, at several levels;

- between the employers and employees,
- between the superiors and subordinates,
- among the colleagues,
- between the producers and the suppliers (spare parts),
- and between the organisation and its customers

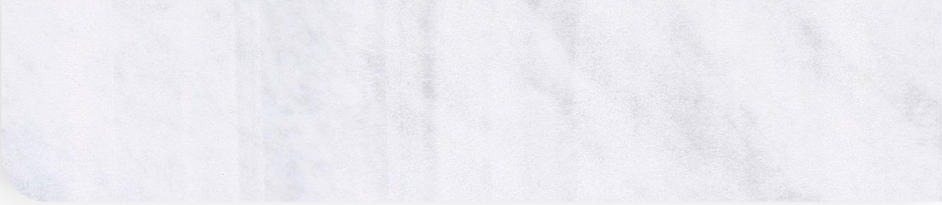


- The absence of cooperation leads to lack of communication, misinformation, void in communication, and undue delay between supply, production, marketing, and consumption.
- This is likely to demoralize and frustrate the employees, leading to collapse of the industry over time and an economic loss to the society.

The impediments to successful cooperation are:

1. Clash of ego of individuals.
2. Lack of leadership and motivation.
3. Conflicts of interests, based on region, religion, language, and caste.
4. Ignorance and lack of interest.





By careful planning, motivation, leadership, fostering and rewarding team work, professionalism and humanism beyond the 'divides', training on appreciation to different cultures, mutual understanding 'cooperation' can be developed and also sustained.

COMMITMENT

- Commitment means alignment to goals and adherence to ethical principles during the activities.
- Holding sustained interest and firmness, in whatever ethical means one follows, with the fervent attitude and hope that one will achieve the goals, is commitment.
- It is the driving force to realize success.
- This is a basic requirement for any profession



- Ex. A design engineer shall exhibit a sense of commitment, to make his product or project designed a beneficial contribution to the society.
- The commitment of top management will naturally lead to committed employees.

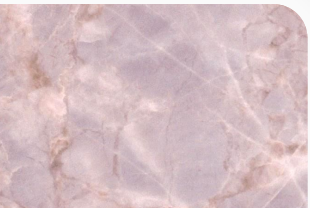
EMPATHY

- Empathy is social radar.
- Sensing what others feel about, without their open talk, is the essence of empathy.
- Empathy begins with showing concern, and then obtaining and understanding the feelings of others, from others' point of view.





- Defined as the ability to put one's self into the psychological frame or reference or point of view of another, to know what the other person feels.
- It includes the imaginative projection into other's feelings and understanding of other's background such as parentage, physical and mental state, economic situation, and association.
- To practice 'Empathy', a leader must have or develop in him, the following characteristics:
 1. Understanding others: sensing others feelings and perspectives, and taking active interest in their welfare.
 2. Service orientation: It is anticipation, recognition and meeting the needs of the clients or customers.

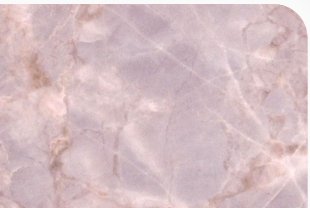


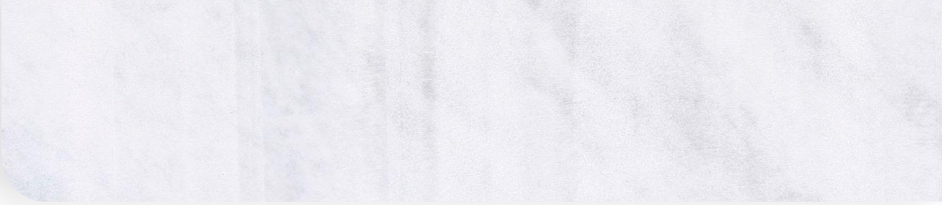


3. Developing others: This means identification of their needs and bolstering their abilities. One should get the feed back, acknowledge the strength and accomplishments, and then coach the individual, by informing about what was wrong, and giving correct feedback and positive expectation of the subject's abilities and the resulting performance.

4. Leveraging diversity (opportunities through diverse people): This leads to enhanced organizational learning, flexibility, and profitability.

5. Political awareness: It is the ability to read political and social currents in an organization.





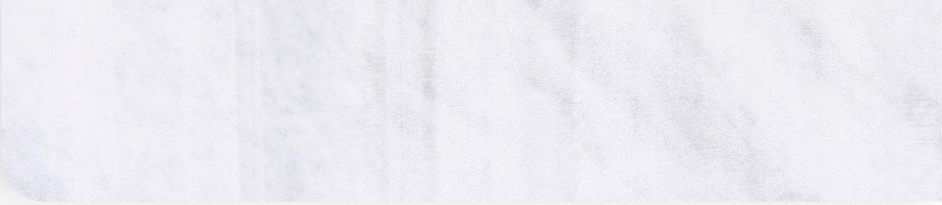
The benefits of empathy include:

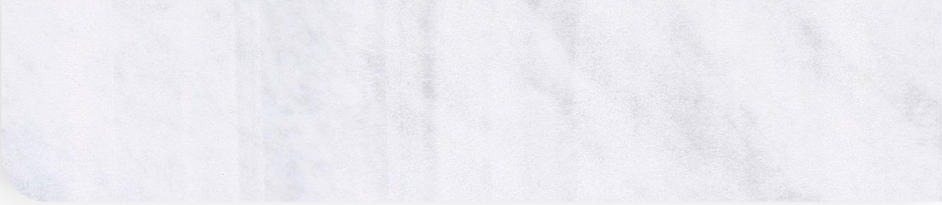
1. Good customer relations (in sales and service, in partnering).
2. Harmonious labor relations (in manufacturing).
3. Good vendor-producer relationship (in partnering.)

SELF-CONFIDENCE

- Certainty in one's own capabilities, values, and goals, is self-confidence.
- These people are usually positive thinking, flexible and willing to change.
- They respect others so much as they respect themselves.



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- Self-confidence is positive attitude, wherein the individual has some positive and realistic view of himself, with respect to the situations in which one gets involved.
 - The people with self-confidence exhibit courage to get into action and unshakable faith in their abilities, whatever may be their positions.
 - They are not influenced by threats or challenges and are prepared to face them and the natural or unexpected consequences.
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The people with self-confidence have the following characteristics:

1. A self-assured standing,
2. Willing to listen to learn from others and adopt (flexibility),
3. Frank to speak the truth, and
4. Respect others' efforts and give due credit.

The factors that shape self-confidence in a person are:

1. Heredity (attitudes of parents) and family environment (elders),
2. Friendship (influence of friends/colleagues),
3. Influence of superiors/role models, and





4. Training in the organization

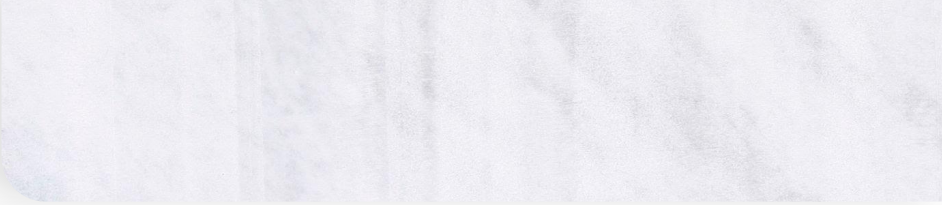
The following methodologies are effective in developing self-confidence in a person:

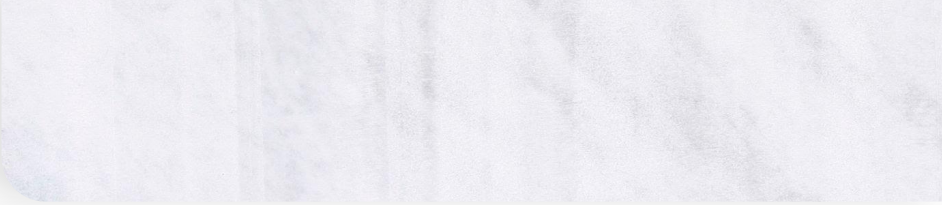
1. Encouraging SWOT analysis. By evaluating their strength and weakness, they can anticipate and be prepared to face the results.
2. Training to evaluate risks and face them (self-acceptance).

SOCIAL EXPECTATIONS

- The norms, rules, and standards of behavior that society or a community expects individuals to follow.



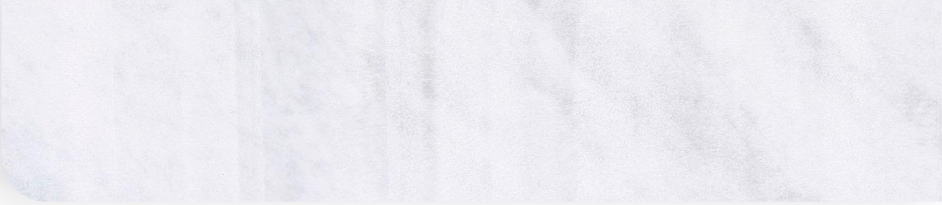
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- They guide how people should act in social, academic, and professional environments, ensuring order, cooperation, and respect within the group.
 - In professional ethics, social expectations mean **acting responsibly, respectfully, and ethically** to maintain trust with society.
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Role of Social Expectations in Professional Ethics

1. **Guides Behavior** → Professionals are expected to uphold honesty, fairness, and responsibility.
2. **Builds Trust** → Society relies on doctors, engineers, teachers, lawyers, etc., to perform duties ethically.
3. **Prevents Misuse of Power** → By following social norms, professionals avoid exploiting their authority.
4. **Promotes Accountability** → Professionals are answerable to both their organizations and the community.
5. **Ensures Social Harmony** → Meeting expectations helps maintain peace and cooperation in society.

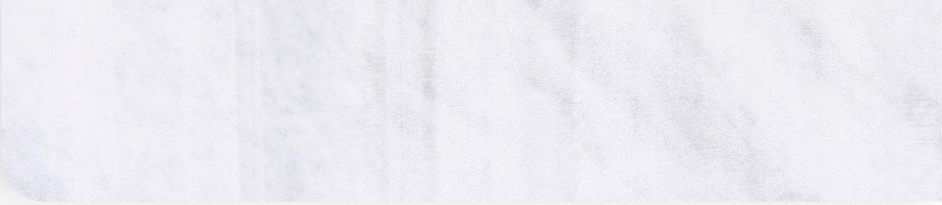


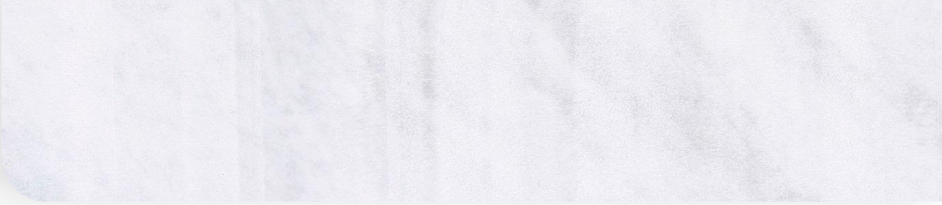


Importance of Social Expectations

- **Maintains Social Order** – Prevents chaos and conflicts by setting acceptable standards.
- **Promotes Respect and Harmony** – Encourages polite, respectful interactions among individuals.
- **Encourages Cooperation** – Helps people work together effectively in families, workplaces, and communities.



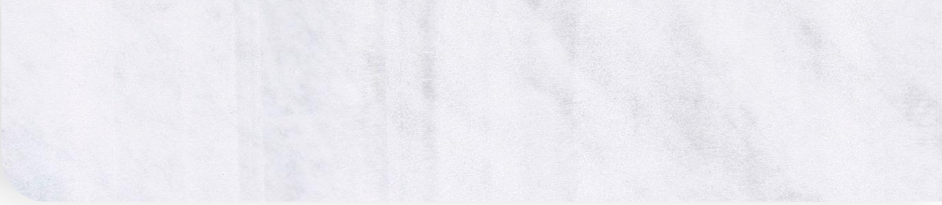
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4. **Reflects Culture and Values** – Different societies have unique customs and traditions that shape expectations.
 5. **Professional Success** – Meeting workplace expectations (like punctuality, teamwork, and honesty) builds trust and reputation.



Examples of Social Expectations in Professions

- **Engineering:** Society expects engineers to design safe, sustainable, and reliable products (e.g., bridges, machines, software).
- **Medicine:** Doctors are expected to prioritize patients' health over personal profit.
- **Teaching:** Teachers are expected to be role models, promoting values like honesty and hard work.
- **Business:** Companies are expected to follow fair trade practices, pay fair wages, and avoid harming the environment.

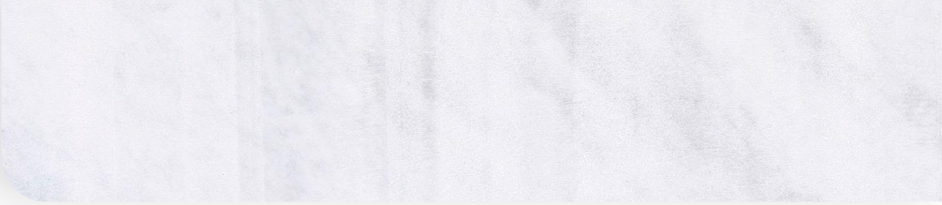




CODE OF CONDUCT IN ENGINEERING PROFESSION

- It is a formal set of principles and rules that govern professional behavior.
- Or it is a **set of ethical and professional standards** that guide engineers in performing their duties responsibly.
- It ensures that engineers uphold **public safety, honesty, fairness, accountability, and sustainability** while practicing their profession.
- It is issued by professional bodies like **IEEE, ASME, Institution of Engineers (India), NSPE (USA)** etc.

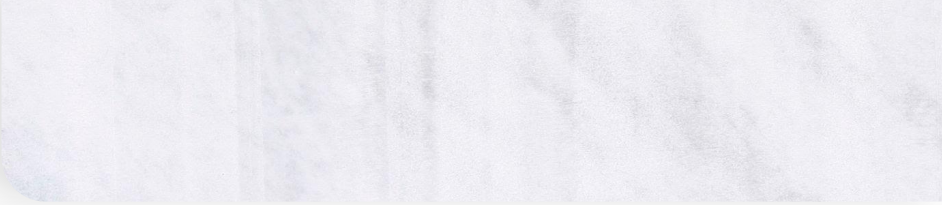




Purpose:

- To **guide engineers** in ethical decision-making.
- To maintain **public trust, safety, and welfare**.
- To ensure **professional integrity and accountability**.

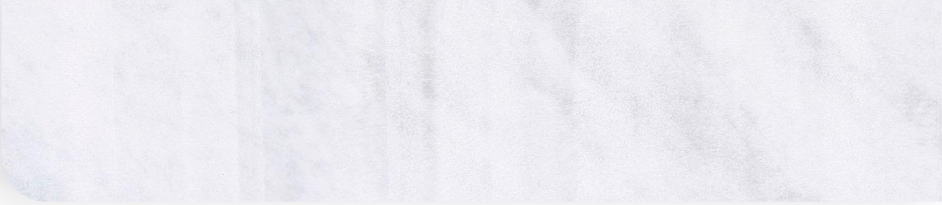




Importance of Code of Conduct

- Maintains **trust** between engineers and society.
- Ensures **safe, reliable, and ethical** engineering practices.
- Prevents **corruption, fraud, and negligence**.
- Enhances the **reputation** of the profession.
- Provides a framework for **decision-making** in ethical dilemmas.





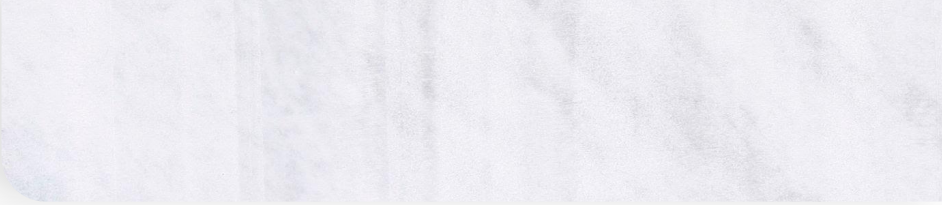
Violation Case:

- *Bhopal Gas Tragedy (1984)* → Poor safety standards and negligence violated engineering ethics, causing thousands of deaths.

Positive Case:

- Engineers working on renewable energy projects (solar/wind power) uphold social and environmental responsibilities.





Key Elements of the Code of Conduct

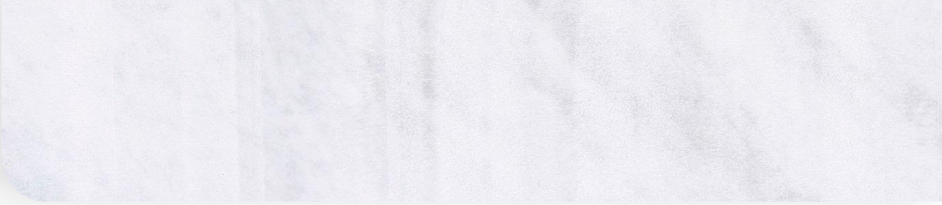
(a) Public Safety and Welfare

- Engineers must hold **paramount the safety, health, and welfare of the public**.
- Ex: An engineer should refuse to approve a bridge design if safety standards are compromised.

(b) Honesty and Integrity

- Present data, reports, and results truthfully.
- Avoid plagiarism, false claims, and misrepresentation of facts.
- Ex: Do not alter test results of materials to make them appear stronger.





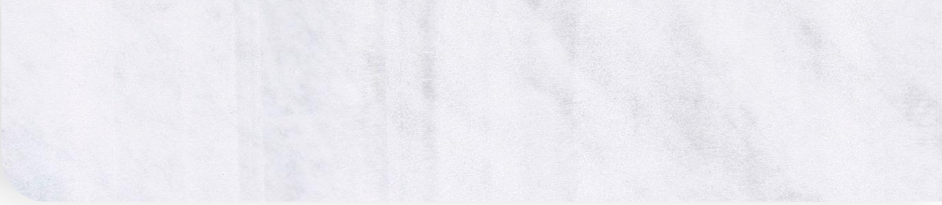
(b) Honesty and Integrity

- Present data, reports, and results truthfully.
- Avoid plagiarism, false claims, and misrepresentation of facts.
- Ex: Do not alter test results of materials to make them appear stronger.

(c) Competence

- Work only in areas of one's competence.
- Continuously upgrade knowledge and skills.
- Ex: A civil engineer should not sign off on an electrical design outside their expertise.





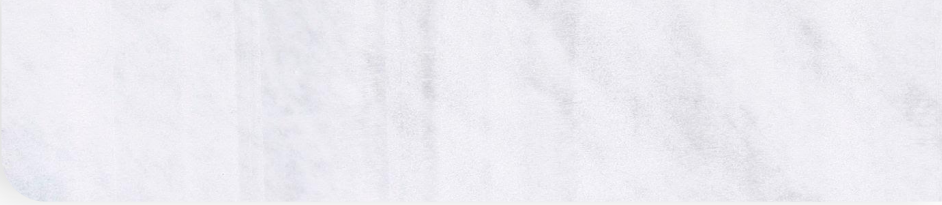
(d) Confidentiality

- Respect the confidentiality of clients, employers, and colleagues.
- Do not misuse sensitive information for personal gain.

(e) Conflict of Interest

- Disclose any personal or financial interest that may affect professional judgment.
- Example: If an engineer owns shares in a construction company, they must declare this before recommending the company for a project.



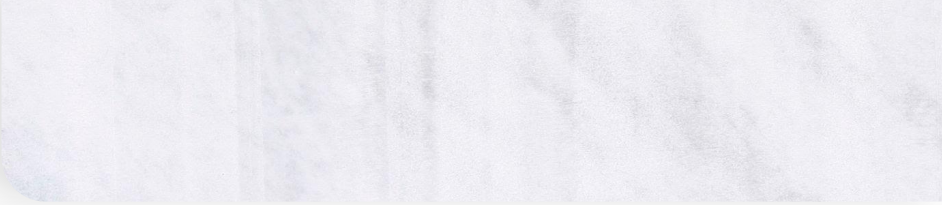


(f) Professional Responsibility

- Report unethical or illegal practices.
- Accept personal accountability for professional decisions.
- Example: If a junior engineer identifies a safety issue, they must escalate it rather than ignore.

(g) Fairness and Respect

- Treat colleagues, clients, and workers with respect, regardless of gender, religion, or background.
 - No discrimination or harassment.
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(h) Environmental Responsibility

- Engineers must ensure their projects do not harm the environment.
- Sustainable design and energy-efficient solutions should be prioritized.

(i) Commitment to Society

- Contribute to society through service, innovation, and ethical practice.
- Example: Designing cost-effective housing for underprivileged communities.

